

Solving Surpassing the Leader

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With optimal play, Hal wins about **51.86%** of games from the 8:12 opening. We compute **9.45 billion reachable state classes** with the leap second available to Baku.

1. The game

In Toshio Sako's *Usogui*, Baku and Hal play Drop the Handkerchief to decide who leads Kakerou. They begin at 8:12 AM with no accumulated time. Hal drops first and Baku checks; they swap roles after each half-round. Both know about the leap second at 8:59:60. Each player chooses a second from 1 to 60 without seeing the other's choice. In the leap window, Baku as Dropper can choose 61. The Checker must still choose from 1 to 60, so a drop at 61 forces a failed check. We solve the public game with this extra action. Hal's memory and the players' private plans are outside this model.

For each player, s is squandered time (ST), the seconds in their cylinder, and t is total time dead (TTD), the sum of prior doses. A check at $c \geq d$ adds $\ell = c - d + 1$ to the Checker's ST; at $s_c + \ell \geq 300$, the Checker loses. A failed check injects $q = s_c + 60$ seconds. The Checker survives with probability

$$p(s, t) = \begin{cases} 0.95(1 - s/240) 0.75^{t/60}, & s \leq 239, s + t \leq 240, \\ 0, & \text{otherwise.} \end{cases}$$

On revival, their ST resets to zero and their TTD gains q . Otherwise, they lose. We use the same survival model as the base game; the manga does not specify a probability formula.

2. Equivalence Class

We keep the role-relative profiles and add identity and time:

$$x = (s_c, t_c, s_d, t_d, h, \tau).$$

Hal drops in half $h = 1$; Baku drops in half $h = 2$. We measure τ in physical seconds from 8:00, including the leap second. The opening has $h = 1$ and $\tau = 720$.

We merge failure-fatal profiles with the same ST, replacing their TTD with $\perp$. This gives the same 17,011 profiles per player as DTH. **Proof.** A successful check changes ST and advances the clock without using TTD. A failed check ends the game when $p = 0$. Thus the merged profiles have the same transitions and payoffs at a fixed half and clock. QED

We group reachable states by half and clock. After Hal revives, we need only 183 Dropper profiles because his ST is zero. We count 9,453,333,117 classes across 2,243 groups.

3. Clock and backwards sweep

Let $T(\tau) = 61$ for $3540 \leq \tau \leq 3600$, and 60 otherwise. We set $q = 0$ after success and use the injected dose after revival. For a live child, define

$$e = \tau + T(\tau) + (q + 120)\mathbf{1}_{q>0}, \quad \tau' = \begin{cases} e + 60, & h = 1, \\ \nu(e), & h = 2. \end{cases}$$

Here $\nu(e)$ is the next minute boundary strictly after e , with 9:00 at $\tau = 3601$. We swap roles and change h . The dose therefore affects when the next player can act.

We sweep clock groups in reverse order, solving each live child before its parent. For $\tau' > 3600$, we read the DTH table: both players have the base game's action sets for the rest of play.

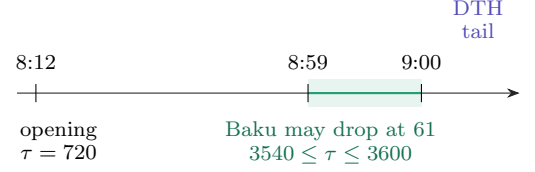


Figure 1: We keep the exact clock until the leap second passes. Both players know the extra action from the opening. The timeline is schematic.

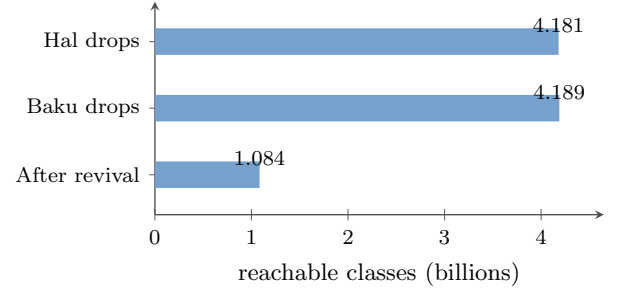


Figure 2: We count each reachable quotient state once. The revival group contains Baku's turns after a survived injection into Hal.

4. The extra second

Let V be the current Dropper's win probability minus loss probability. For the success child x_ℓ and revival child x_f , set

$$S_\ell = \begin{cases} 1, & s_c + \ell \geq 300, \\ -V(x_\ell), & \text{otherwise,} \end{cases}$$

$$F = \begin{cases} 1, & p = 0, \\ 1 - p - pV(x_f), & p > 0. \end{cases}$$

For the ordinary seconds, we solve

$$M_{d,c} = \begin{cases} S_{c-d+1}, & c \geq d, \\ F, & c < d. \end{cases}$$

Successes share one child clock; failures share another for the given Checker profile. We still need only 60 success payoffs and F . In Baku's window, we add a row of 60 copies of F . If v_{60} is the square matrix value, then

$$v_{61} = \max\{v_{60}, F\}.$$

Proof. Give the extra row mass α and an ordinary Dropper mixture a the remaining mass. The guaranteed payoff is $\alpha F + (1 - \alpha) \min_c (a^\top M)_c$. We maximize over a , then choose the better endpoint in α . The Checker can use a minimizing mixture for M in either case. QED

5. Computing both players' strategies

We first test fixed seconds with a row-minimum and column-maximum scan. For a mixed strategy, put $D = S_1 - F$ and $\Delta_m = S_{m+1} - S_m$. For $D \neq 0$, compute

$$r_1 = 1, \quad r_k = -\frac{1}{D} \sum_{m=1}^{k-1} \Delta_m r_{k-m} \quad (2 \leq k \leq 60).$$

For finite nonnegative weights, normalize with $W = \sum_k r_k$: $a_k = r_k/W$ and $b_k = a_{61-k}$. **Proof.** Subtract adjacent row payoffs:

$$(Mb)_d - (Mb)_{d+1} = Db_d + \sum_{m=1}^{60-d} \Delta_m b_{d+m}.$$

Setting these differences to zero gives the recurrence in reverse order. Column differences give the same weights for the Dropper. Both players make their opponent indifferent between seconds. QED We compute the value as $F + D/W$ when the nonnegative weights meet the rounding-bound assumptions. We enclose both saddle bounds within 10^{-10} of this center. For clipped weights, we check the matrix. The leap continuation changes the payoffs, so the recurrence need not produce an accepted pair at each state.

6. Packing and error bounds

For an unresolved stage with $\delta = F - S_1 > 0$, define

$$A_{ij} = \begin{cases} (F - S_{j-i+1})/\delta, & j \geq i, \\ 0, & j < i. \end{cases}$$

Then $M = F\mathbf{1}\mathbf{1}^\top - \delta A$, with unit diagonal in A . We solve the dual linear programs

$$\max_{y \geq 0} \{\mathbf{1}^\top y : A^\top y \leq \mathbf{1}\} = \min_{z \geq 0} \{\mathbf{1}^\top z : Az \geq \mathbf{1}\} = Z.$$

Proof. The unit diagonal gives a finite nonnegative covering vector by back substitution; zero is feasible for packing. Linear-programming duality gives a common optimum $Z > 0$. For $a = y/Z$ and $b = z/Z$, we obtain $a^\top M \geq (F - \delta/Z)\mathbf{1}^\top$ and $Mb \leq (F - \delta/Z)\mathbf{1}$. Thus $v_{60} = F - \delta/Z$. QED

We use a Rust simplex kernel with at most 512 pivots per candidate. We send unsupported or rejected stages to HiGHS. For a candidate pair, we check the original matrix with

$$L = \min_c (a^\top M)_c, \quad U = \max_d (Mb)_d.$$

We require $U - L \leq 10^{-6}$. In the window, we replace each bound by its maximum with F . We store the midpoint or a center with proved error at most 5×10^{-7} .

Accumulated bound. The DTH progress score $\Phi = s_c + s_d + \rho_c + \rho_d$, with $\rho = t$ for $p > 0$ and 301 otherwise, increases on each live transition. The clock and the extra failure row preserve that property. A child-value error changes a stage value by at most the same amount. Using the same local bound in the DTH tail, induction gives

$$|\hat{V} - V| \leq (1201 - \Phi)5 \times 10^{-7}.$$

At the opening, the win-probability error is at most 0.000301. We use floating-point saddle checks for the matrix and LP paths; the sampled audit measures Bellman residuals against stored children.

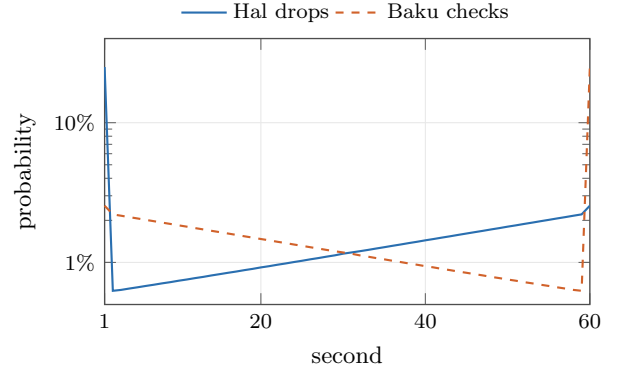


Figure 3: At the opening, Hal puts 25.01% on second 1 and Baku puts 25.01% on second 60. We obtain both mixtures from an LP check of the opening matrix.

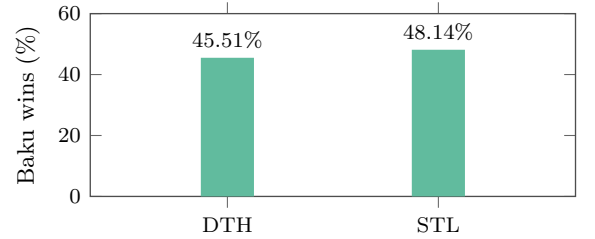


Figure 4: Baku gains 2.63 percentage points over the base game from the same zero-load opening, with Hal dropping first.

7. Results

At the 8:12 opening, we obtain

$$V_{\text{Hal}} = 0.03717 \pm 0.00061, \\ P(\text{Hal wins}) = 0.51859 \pm 0.00031.$$

We accept 8,843,634,465 states in the fixed-action and recurrence kernel. We resolve the remaining 609,698,652 through 306,553,687 native packing certificates, 221,618,372 HiGHS solves, and 81,526,593 reuses of certified revival values. We check packed membership and finite values for all 9,453,333,117 reachable states. We re-solve 4,621 sampled stages with LP. The largest Bellman residual is 3.07×10^{-9} ; the largest sampled saddle gap is 1.02×10^{-9} . At the opening, we measure a gap of 1.58×10^{-15} .

We retain values in packed arrays with a reachability bitmap and hashes for the source and DTH tail. The completed run records 12,820.8 seconds across resumed work and 13.14 GB peak resident memory. Some retained LP work predates the component timers; we report the saved run's elapsed total.

Sources. T. Sako, *Usogui*, Shueisha, 2006–2018. J. Cenawood, *Solving Drop the Handkerchief*, 2026. Public-game clock and survival contracts; completed **leap-full-native** report and packed-coverage audit, September 2026.